

DHRUV SOOD

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TECHNICAL SKILLS

LLM Systems: LangGraph, LangChain, MCP, Azure OpenAI, RAG, Multi-Agent Systems, Text2SQL, Tool Calling, Guardrails, Evals
ML & Data Science: pandas, scikit-learn, LightGBM, XGBoost, PyTorch, Forecasting, Causal Inference (DiD), Fine-tuning
Data & Retrieval: PostgreSQL, Redis, OpenSearch, Elasticsearch, Qdrant, Hybrid Search, Apache Airflow, Amazon Redshift
Cloud & Infra: Azure (AKS, OpenAI, Functions), AWS, Docker, Kubernetes, FastAPI, Streamlit, Datadog, pytest, CI/CD
Programming: Python, SQL, TypeScript, C++

EXPERIENCE

Rocketfrog.ai

Remote

AI Engineer

Dec 2025 – Present

Client-facing AI consultancy. Delivered production GenAI and ML systems for Sonny's Enterprises and WashMetrix/Quivio.
WashMetrix/Quivio — AI analytics chatbot *LangGraph, Azure OpenAI, MCP, FastAPI, Redis, AKS, Datadog*

- Built a **multi-tenant LLM agent** answering business questions with text, charts and tables across **31 sales, labor, membership and accounting KPIs**; served via FastAPI on **AKS** with Redis-checkpointed state across replicas.
- Split data access into a standalone **MCP service** (streamable HTTP); passed tenant scope as request metadata the model cannot alter, with per-request isolation and ownership checks on cached payloads to **prevent cross-tenant leaks**.
- **Cut LLM cost and latency** with a two-tier input guardrail (regex fast path, then LLM scope classifier), a conditional output guardrail and period-aware Redis caching; sandboxed LLM-generated code with a timeout and import allowlist.
- Enforced **schema-validated structured output** and enum-typed tool parameters to eliminate invalid model calls; instrumented with **Datadog LLM Observability** and cross-service trace IDs.

Sonny's Enterprises — Demand-forecasting API

XGBoost, LightGBM, scikit-learn, Airflow, PostgreSQL, pytest

- Built and shipped the **hourly demand-forecasting layer** behind Sonny's customer-facing forecast API, delivering 15-day hourly predictions for **218 car-wash sites** (360 forecasts per site per day).
- **Cut MAE 41%** (11.35 → 6.70) and WMAPE 41.5% → 24.5% via head-to-head evaluation on 297K site-hours; shipped the winning XGBoost + LightGBM engine behind a config switch, **shadow-running** it in production against the incumbent scikit-learn suite.
- Migrated a cron-driven ML pipeline to **Airflow** (feature-store and prediction DAGs, cross-DAG sensors, alerting) and designed its **8.2M-row Postgres prediction warehouse**, halving storage by right-typing columns; backfilled 106 origins via idempotent upserts.
- Removed a **silent failure** dropping 158 of 218 sites with a fail-fast coverage guard; root-caused two production defects (duplicate site-hours across 2,870 record groups, a serving regression collapsing 28-day trends) and added ~50 tests incl. 1e-5 parity checks.
- *Sonny's Enterprises — Site-selection platform* *LightGBM, scikit-learn, LangGraph, GPT-4o, FastAPI, Streamlit*
- Built a **site-selection platform** forecasting 5-year volume and P&L for any US location (71.7K site-months); LightGBM/ExtraTrees ensemble with opening-ramp and competitor models **halved legacy proforma error** (29% vs 58% median, p<0.001).
- Designed a **council of 5 LLM agents** (LangGraph, GPT-4o, cited web search) recommending whether to build; backtested on 420 historical builds at **52% precision vs a 43% base rate**.
- Ran **causal analyses** (difference-in-differences, placebo tests, 751 competitor openings) that **changed site-selection criteria**; showed 31 demographic measures had no predictive power and cut a claimed campaign lift from +15.9% to +7.1%.

Artificial Intelligence Institute of South Carolina

Remote

AI Research Intern

Sep 2025 – Dec 2025

- Researched LLM bias, safety, and **constitutional alignment frameworks** using curated prompt sets, fairness-aware decoding experiments, and cross-model benchmarking across LLMs, achieving up to **26% reduction in identity bias** at inference time.
- **Preprint:** AMBEDKAR — Fairness-Aware Decoding for Constitutional Alignment.

Probo (Operations Discontinued)

Gurgaon, India

Software Engineer

Jan 2025 – Aug 2025

- Led development of a **production-grade multi-agent RAG system** on OpenSearch, built on a **text-to-SQL** retrieval pipeline with structured tool calling; balanced quality, latency and inference cost under live traffic, driving a **35% increase in user engagement**.

National Stock Exchange (NSE India)

New Delhi, India

ML Engineering Intern

May 2024 – Jul 2024

- Architected and deployed a **RAG-based regulatory intelligence system** over 10K documents on Qdrant using hybrid search.

EDUCATION & ACHIEVEMENTS

Indraprastha Institute of Information Technology (IIIT-Delhi)

New Delhi, India

Bachelor of Technology in Computer Science & Social Sciences (8.04/10)

Nov 2021 – May 2025

- **Coursework:** Machine Learning, NLP, Information Retrieval, Databases, Data Structures & Algorithms, AI Governance
- **Achievements:** State Topper (NGSE); Top 5% Adobe GenSolve'24; 2nd place IEEE BlackSlash
- **Leadership:** Academic mentor; open-source contributions; technology education initiatives